Alcohol Consumption does not have a significant impact on life Expectany.

Exploring the Impact of Alcohol Consumption on Life Expectancy in South and Southeast Asia

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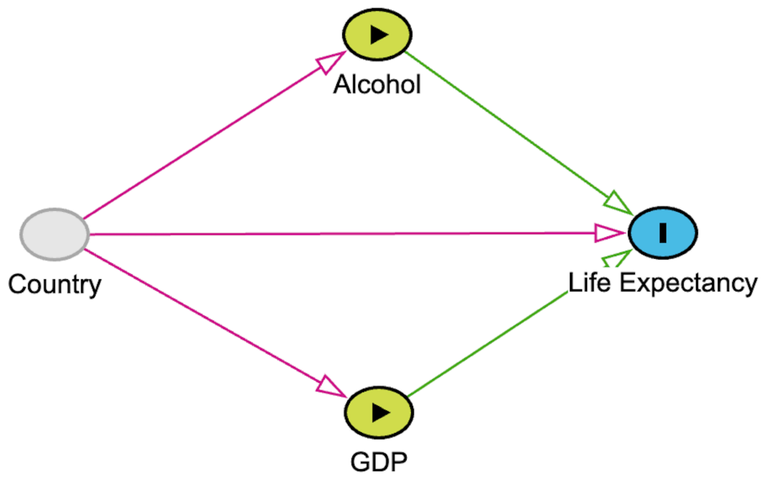
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Introduction

Alcohol consumption is a major public health issue, contributing to over 3 million deaths annually, or 5.3% of global deaths, with long-term use linked to chronic diseases that reduce life expectancy. In South and Southeast Asia, alcohol consumption varies widely, with some countries having minimal or banned consumption (e.g., Afghanistan, Pakistan) and others (e.g., India, Thailand) exceeding 1.0 liter per capita. High consumption is associated with poor health outcomes, particularly in developing countries with underdeveloped healthcare systems. While countries with higher healthcare spending tend to have better life expectancy, many in this region prioritize economic development over health, worsening the effects of alcohol misuse. This research examines the impact of alcohol consumption on life expectancy in the region, based also on how much GDP is contributed, while focusing on countries with low and high alcohol consumption.

Model

To identify the causal effect of GDP, alcohol consumption, and region on life expectancy, I include GDP (log-transformed), alcohol consumption, region, and their interaction terms as explanatory variables. The log transformation of GDP accounts for diminishing returns, while the interaction terms assess whether the effects of GDP and alcohol differ between South and Southeast Asia. The regression results show that GDP has a positive effect on life expectancy, with a stronger relationship observed in Southeast Asia compared to South Asia. The significant interaction term indicates that GDP growth leads to larger improvements in life expectancy in Southeast Asia. Additionally, alcohol consumption shows a weak and inconsistent relationship with life expectancy, suggesting that other regional factors play a more significant role in determining health outcomes.



Data

For the purpose of this research, I used data from the World Health Organization (WHO) which I found on Kaggle, which provides comprehensive global health-related statistics and indicators compiled from recognized international sources. The dataset focuses on countries in South Asia and Southeast Asia from 2000-2015, two regions that exhibit significant diversity in socioeconomic and health-related outcomes. For the purposes of this study, I primarily utilized the variables Log(GDP), Alcohol(Liter in pure Alc), Development Status (Developing, Developed), and Region to explore how life expectancy trends and economic outcomes differ between South Asia and Southeast Asia. By separating countries by regions and development status, this research seeks to understand the relationship between alcohol consumption, and regional disparities.

Results

The regression analysis examines the effect of minimal alcohol consumption (<1.0 liters) on life expectancy in South and Southeast Asia. The model shows an R^2 of near zero (0.000067), meaning alcohol consumption explains almost no variation. The slope (1.7754) indicates a small, statistically insignificant positive relationship (p = 0.942) with a large standard error. The F-statistic (0.00526) confirms no explanatory power. The weak, slightly positive regression line suggests minimal impact of alcohol on life expectancy in these lower-middle-income countries.

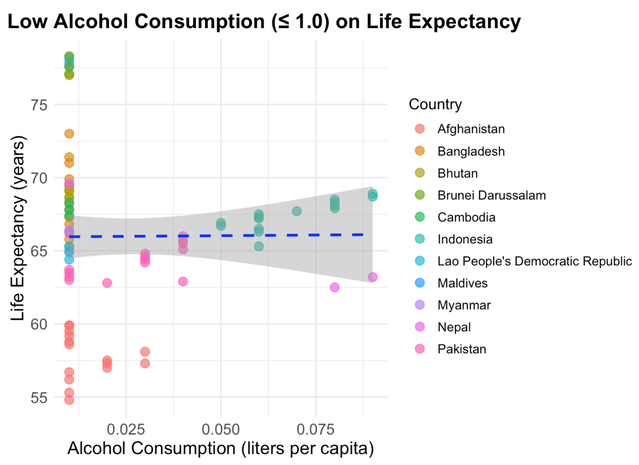


Figure 1: Countries with alcohol consumption 0 to 1.0 liters per capita

The second analysis examines the relationship between alcohol consumption and life expectancy with countries having more than 1.0 liters per capita, this also mentions countries with a higher GDP overall. The results show a weak negative association, with the regression equation: Life Expectancy = 73.01 - 0.6464 × Alcohol Consumption. The slope (-0.6464) indicates that for each additional liter of alcohol consumed per capita, life expectancy decreases slightly. However, the effect is not statistically significant at the 95% confidence level (p = 0.0688). The model explains only 2.67% of the variation in life expectancy (R^2 = 0.0267), suggesting that alcohol consumption has minimal explanatory power. The graph further illustrates this weak relationship, with considerable variability and wide confidence intervals, indicating that other factors likely play a more significant role in determining life expectancy. The shaded confidence band around the regression line in the graph further illustrates the high uncertainty, particularly for higher alcohol consumption levels. Countries such as Singapore and Thailand (colored blue and purple, respectively) exhibit relatively high life expectancy despite moderate alcohol consumption, while others like India and Cambodia (orange and pink) display lower life expectancy at similar alcohol levels

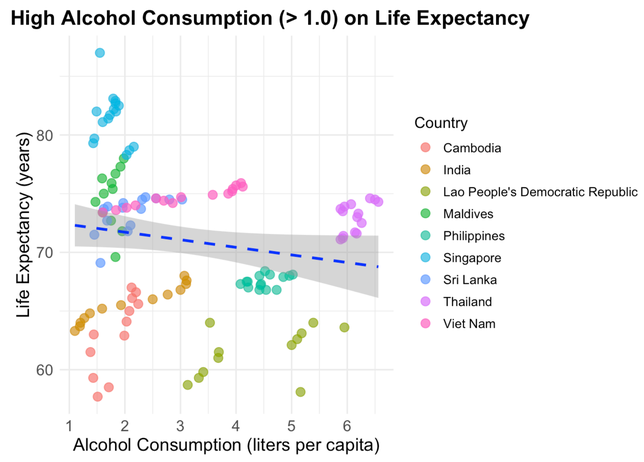


Figure 2: Countries with alcohol consumption > 1.0 liters per capita

Discussion

The objective of this analysis was to examine the relationship between alcohol consumption and life expectancy in South and Southeast Asian countries. The regression results revealed no significant relationship, as the near-zero R-squared value (0.000067) and the high p-value (0.942) indicate that alcohol consumption does not meaningfully explain variations in life expectancy. While this result does not support a direct relationship, it highlights the likelihood of other factors, such as acess to healthcare expenditure, or pollution, being more influential in determining life expectancy. The findings appear reasonable given that life expectancy is influenced by a complex interplay of economic, social, and health factors beyond alcohol consumption alone. However, this study has limitations, including a small sample size, limited control variables, and potential data accuracy issues. For future research, incorporating additional variables such as healthcare quality and pollution metrics could provide a more comprehensive understanding of life expectancy determinants.

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